

2.8% (std. err. 2.9%), which is positive but statistically insignificant. We can also divide the employment change $\Delta a + \Delta b$ by the sample-averaged minimum wage increase of 10.1% to calculate the employment elasticity with respect to the minimum wage of 0.024 (std. err. 0.025). This estimate is statistically insignificant, and the 95% confidence interval rules out substantial reductions in the aggregate employment, including the baseline aggregate employment elasticity of -0.074 in [Meer and West \(2016\)](#) (see their Table 4). The most common minimum wage employment elasticities are from teens; for example, [Neumark and Wascher \(2008\)](#) argue that this falls between -0.1 and -0.3 , while [Allegretto et al. \(2017\)](#) argue that it is closer to 0. However, the directly affected share of teens (43.2%) is much larger than the workforce overall (8.6%). Therefore, to make our estimates on overall employment comparable to the estimates for teens we can multiply our estimate and standard errors by the ratio of the shares $\frac{0.432}{0.086} = 5.02$. This leads to an affected-share-adjusted 95% confidence interval of $[-0.13, 0.37]$, which rules out most of the -0.1 to -0.3 range.

Second, using the formula in [equation \(2\)](#), we can also calculate the change in the average wage and the employment elasticity with respect to own wage (i.e., the labor demand elasticity in the competitive model). We estimate that the effect of the minimum wage on average wages is 6.8% (std. err. 1.0%), which is statistically significant. The estimate for the elasticity of employment with respect to own wage is 0.411 (std. err. 0.430). The confidence intervals rule out any own-wage elasticities more negative than -0.450 at the 95% confidence level. Such a lower bound rules out many estimates in the literature that found a negative employment elasticity (see [Online Appendix Figure A.7](#); also, [Neumark and Wascher \(2008\)](#) argue that the own-wage employment elasticity can easily be -1 or even -2).

[Figure III](#) shows the changes in the missing jobs paying below the new minimum wage (Δb_τ), and the excess jobs paying up to \$4 above the minimum wage (Δa_τ) over annualized event time using our baseline specification. All the estimates are expressed as changes from event date $\tau = -1$, or the year just prior to treatment, the estimates for which are normalized to 0. There are four important findings that we would like to highlight. First, we find a very clear reduction in the jobs paying below the new minimum wage (shown in red dashed line) between the year just prior to treatment ($\tau = -1$) and the year of treatment ($\tau = 0$)—this

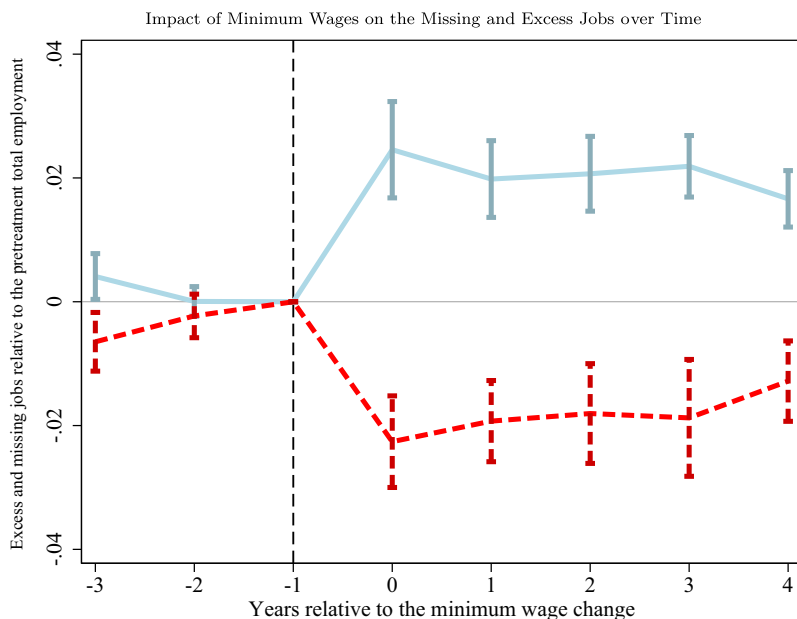


FIGURE III

Impact of Minimum Wages on the Missing and Excess Jobs over Time

The figure shows the main results from our event study analysis (see [equation \(1\)](#)) exploiting 138 state-level minimum wage changes between 1979 and 2016. The figure shows the effect of a minimum wage increase on the missing jobs below the new minimum wage (red dashed line; color version available online) and on the excess jobs at or slightly above it (blue solid line) over time. The red dashed line shows the evolution of the number of jobs (relative to the total employment one year before the treatment) between \$4 below the new minimum wage and the new minimum wage (Δb); the blue solid line shows the number of jobs between the new minimum wage and \$5 above it (Δa). We also show the 95% confidence interval based on standard errors that are clustered at the state level.

shows that the minimum wage increases under study are measurably binding. Second, although there is some reduction in the magnitude of the missing jobs in the post-treatment window, it continues to be very substantial and statistically significant five years out, showing that the treatments are fairly durable over the medium run.²⁰ Third, the response of the excess jobs at or above the new minimum (Δa) exhibits a very similar pattern in magnitudes, with the opposite sign. There is an unmistakable

20. The durability of the treatment can also be seen in [Online Appendix Figure A.4](#), which plots the progression of the minimum wage using our event study design.

jump in excess employment at $\tau = 0$, and a substantial portion of it persists and is statistically significant even five years out. Fourth, for both the changes in the excess and missing jobs there is only a slight indication of a preexisting trend prior to treatment. The $\tau = -2$ leads are statistically indistinguishable from 0 and although there is some evidence of changes three years prior to treatment, the leading effects are very small relative to the post-treatment effect estimates. Moreover, the slight downward trend in excess jobs, and the slight upward trend in missing jobs is consistent with a falling value of the real minimum wage prior to treatment. The sharp jump in both the excess and missing jobs at $\tau = 0$, the lack of substantial pretreatment trends, and the persistent post-treatment gap between the two shares all provide strong validation of the research design. [Online Appendix Figure A.5](#) shows analogous time paths for wages and employment showing a sharp and persistent wage effect at $\tau = 0$ coupled with little change in employment over the event window—either before or after treatment.

1. Robustness Checks. In [Table I](#), we assess the robustness of the main results to including additional controls for time-varying, unobserved heterogeneity. This is particularly important because results in the existing literature are often sensitive to the inclusion of various versions of time-varying heterogeneity (e.g., [Neumark, Salas, and Wascher 2014](#), [Allegretto et al. 2017](#)). In column (1) we report the five-year-averaged post-treatment estimates for the baseline specification shown in [Figures II and III](#). Columns (2) and (3) add wage-bin-by-state specific linear and quadratic time trends, respectively. Note that in the presence of three pretreatment and five post-treatment dummies, the trends are estimated using variation outside of the eight-year window around the treatment, and thereby are unlikely to be affected by either lagged or anticipation effects. Columns (4)–(6) allow the wage-bin-period effects to vary by the nine census divisions. Column (6) represents a highly saturated model allowing for state-specific quadratic time trends and division-period effects for each \$0.25 wage bin.

Overall, the estimates from the additional specifications are fairly similar to the baseline estimate. In all cases, there is a clear bite of the policy as measured by the reduction in jobs paying below the minimum, Δb . Consistent with the presence of a substantial bite, there is a statistically significant increase in real wages of affected workers in all specifications: these range

TABLE I
IMPACT OF MINIMUM WAGES ON EMPLOYMENT AND WAGES

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Missing jobs below new MW (Δb)	-0.018*** (0.004)	-0.018*** (0.004)	-0.018*** (0.004)	-0.016*** (0.002)	-0.016*** (0.002)	-0.015*** (0.002)	
Excess jobs above new MW (Δa)	0.021*** (0.003)	0.018*** (0.003)	0.020*** (0.003)	0.016*** (0.002)	0.014*** (0.003)	0.015*** (0.003)	
% Δ affected wages	0.068*** (0.010)	0.057*** (0.010)	0.068*** (0.012)	0.049*** (0.010)	0.043*** (0.010)	0.050*** (0.011)	0.065*** (0.010)
% Δ affected employment	0.028 (0.029)	0.000 (0.023)	0.022 (0.021)	-0.002 (0.021)	-0.019 (0.021)	-0.000 (0.023)	0.027 (0.028)
Employment elasticity w.r.t. MW	0.024 (0.025)	0.000 (0.020)	0.019 (0.018)	-0.001 (0.018)	-0.016 (0.018)	-0.000 (0.019)	0.023 (0.024)
Emp. elasticity w.r.t. affected wage	0.411 (0.430)	0.006 (0.402)	0.326 (0.313)	-0.032 (0.439)	-0.449 (0.574)	-0.003 (0.455)	0.410 (0.421)
Jobs below new MW ($\bar{b}_{-1}$)	0.086	0.086	0.086	0.086	0.086	0.086	0.086
% Δ MW	0.101	0.101	0.101	0.101	0.101	0.101	0.101
Number of events	138	138	138	138	138	138	138
Number of observations	847,314	847,314	847,314	847,314	847,314	847,314	14,484
Number of workers in the sample	4,694,104	4,694,104	4,694,104	4,694,104	4,694,104	4,694,104	4,694,104